THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 1 (Suggested Solution)

1. **(Bookwork)**

   (a) See *p.11 of Chapter 1*:
   Let $A \subseteq \mathbb{R}$ be a nonempty set which is bounded above. Then $\sup A$ exists in $\mathbb{R}$.

   (b) See *Theorem 1.38 of Chapter 1*:
   For each $n \in \mathbb{N}$, if $I_n = [a_n, b_n]$ is a nested sequence of closed and bounded intervals, then there exists $x \in \mathbb{R}$ such that $x \in I_n$ for all $n \in \mathbb{N}$, that is, we have

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

   (c) See *Theorem 1.38 of Chapter 1*:
   Since $I_n = [a_n, b_n]$ is a nested sequence of intervals, we have

$$a_1 \leq a_2 \leq a_3 \leq \cdots \leq a_n \leq \cdots\cdots \leq b_n \leq \cdots \leq b_3 \leq b_2 \leq b_1.$$

   Consider the set $A = \{a_n : n \in \mathbb{N}\}$ of left endpoints of the intervals. Thus each b_n is an upper bound for A. By the Completeness Axiom, the supremum $x = \sup A$ exists. We now claim that $x \in I_n$ for all $n \in \mathbb{N}$. Consider a particular $I_n = [a_n, b_n]$. Since x is an upper bound for A, we have $a_n \leq x$. But b_n is an upper bound for A and x is the least upper bound. Thus we have $x \leq b_n$. Hence $x \in I_n$ and since this is true for every $n \in \mathbb{N}$, we conclude that $x \in \bigcap_{n=1}^{\infty} I_n$ and the intersection is nonempty.

2. (a) (**Bookwork** – see *Assignment 2, Q.19*)
   The given condition implies that

$$|x_{n+2} - x_{n+1}| \leq \alpha^n |x_2 - x_1| \quad \text{for all } n \in \mathbb{N}.$$

   Note that if $x_1 = x_2$, then the above suggests that $\{x_n\}$ is a constant sequence and hence must be a Cauchy sequence. Thus we may assume that $x_1 \neq x_2$.
   If $n > m \geq 2$, then

$$\begin{aligned}
|x_n - x_m| &= |(x_n - x_{n-1}) + (x_{n-1} - x_{n-2}) + \cdots + (x_{m+1} - x_m)| \\
&\leq |x_n - x_{n-1}| + |x_{n-1} - x_{n-2}| + \cdots + |x_{m+1} - x_m| \\
&\leq (\alpha^{n-2} + \cdots + \alpha^{m-1}) |x_2 - x_1| \\
&= \alpha^{m-1} \left(\frac{1 - \alpha^{n-m}}{1 - \alpha} \right) |x_2 - x_1| \\
&< \alpha^{m-1} \left(\frac{1}{1 - \alpha} \right) |x_2 - x_1| \;=\; C\alpha^{m-1},
\end{aligned}$$

where $C = \left(\frac{1}{1-\alpha}\right)|x_2 - x_1|$ is a positive constant. Now since $0 < \alpha < 1$, the sequence $\{\alpha^m\}$ converges to 0. Thus for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $m \in \mathbb{N}$, if $m \geq N$,

$$\alpha^m < \frac{\alpha}{C}\,\epsilon, \quad \text{or equivalently,} \quad \alpha^{m-1} < \frac{1}{C}\,\epsilon.$$

It follows that for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $n, m \in \mathbb{N}$, if $n > m \geq N$, then $|x_n - x_m| < \epsilon$.

(b) First of all, note that

$$|x_{n+2} - x_{n+1}| = \left|\frac{1}{6}\left(x_{n+1}^2 + 8\right) - \frac{1}{6}\left(x_n^2 + 8\right)\right| = \frac{1}{6}|x_{n+1} + x_n| \cdot |x_{n+1} - x_n|. \qquad (*)$$

We then claim that $0 < x_n < 2$ for any $n \in \mathbb{N}$. We shall prove this by induction. Clearly we have $0 < x_1 = 1 < 2$. For any integer $k \geq 1$, assume that $0 < x_k < 2$. Then it is easy to see that $x_{k+1} > 0$. Moreover,

$$\begin{aligned} x_{k+1} \;=\; \frac{1}{6}\left(x_k^2 + 8\right) \;&<\; \frac{1}{6}\left(2^2 + 8\right) \quad (\text{since } 0 < x_k < 2) \\ &=\; 2. \end{aligned}$$

Thus, by induction, we have $0 < x_n < 2$ for any $n \in \mathbb{N}$. Using this, $(*)$ becomes

$$|x_{n+2} - x_{n+1}| \leq \frac{1}{6}(|x_{n+1}| + |x_n|) \cdot |x_{n+1} - x_n| < \frac{4}{6}|x_{n+1} - x_n| = \frac{2}{3}|x_{n+1} - x_n|.$$

By the result of (a), $\{x_n\}$ is a Cauchy sequence (and hence converges to some L). Thus

$$L \;=\; \lim_{n\to\infty} x_{n+1} \;=\; \lim_{n\to\infty} \frac{1}{6}\left(x_n^2 + 8\right) \;=\; \frac{1}{6}\left(\lim_{n\to\infty} x_n^2 + 8\right) \;=\; \frac{1}{6}\left(L^2 + 8\right),$$

which means $L^2 - 6L + 8 = 0$ and hence $L = 2$ or 4. Since $x_n < 2$ for all $n \in \mathbb{N}$, we must have $L \leq 2$. It follows that $L = 2$.

3. (a) Firstly, since $\alpha\beta = k > 0$ and $\alpha + \beta = 1$, so we know that both $\alpha, \beta > 0$. Then we apply mathematical induction. The basis case is true as we are given that $\alpha < a_1 = h < \beta$. For any integer $m \geq 1$, assume that $\alpha < a_m < \beta$. Then we have $\alpha^2 < a_m^2 < \beta^2$ since $\alpha, a_m, \beta > 0$. So

$$\alpha = \alpha^2 + k < a_m^2 + k < \beta^2 + k = \beta \quad (\text{as } \alpha, \beta \text{ are the roots of the given equation})$$

and hence $\alpha < a_{m+1} < \beta$. Thus, by induction, $\alpha < a_n < \beta$ is true for any $n \in \mathbb{N}$.

(b) Note that for any $x \in \mathbb{R}$,

$$\alpha < x < \beta \iff x^2 - x + k < 0.$$

It follows from (a) that $a_n^2 - a_n + k < 0$ for all $n \in \mathbb{N}$, i.e. $a_{n+1} - a_n < 0$, and hence $\{a_n\}$ is decreasing.

(c) Note that $\{a_n\}$ is decreasing and bounded below by α, so $\{a_n\}$ converges by the Monotone Convergence Theorem. Suppose $\{a_n\}$ converges to L. Then we have

$$\begin{aligned} & \lim_{n\to\infty} a_{n+1} \;=\; \lim_{n\to\infty}\left(a_n^2 + k\right) \\ \Longrightarrow\quad & \lim_{n\to\infty} a_{n+1} \;=\; \left(\lim_{n\to\infty} a_n\right)^2 + k \\ \Longrightarrow\quad & L \;=\; L^2 + k \\ \Longrightarrow\quad & L = \alpha \quad \text{or} \quad L = \beta. \end{aligned}$$

Since $\{a_n\}$ is decreasing and $\beta > a_1$, so L cannot be β and hence $\{a_n\}$ converges to α.

4. (a) True

First of all, it is clear that A is a nonempty subset of $\mathbb{R}$ and 0 is a lower bound of A. Hence, by the Completeness Axiom, $\inf A$ exists. We shall apply *Theorem 1.28* to show that $\inf A = 0$.

For any $\epsilon > 0$, by the Archimedean Property, there exists $N \in \mathbb{N}$ such that $\frac{1}{N} < \epsilon$. Now consider an element

$$a = \frac{1}{2N} + \frac{1}{2^{2N}} \in A \quad \text{(take } n = m = 2N\text{)}$$

Then this a satisfies

$$a = \frac{1}{2N} + \frac{1}{2^{2N}} < \frac{1}{2N} + \frac{1}{2N} = \frac{1}{N} < \epsilon.$$

(Here we have used that fact that $2^k > k$ for any $k \in \mathbb{N}$). Hence we have proved that for any $\epsilon > 0$, there exists $a \in A$ such that $a < \epsilon$. Since 0 is also a lower bound of A, we conclude that $\inf A = 0$.

(b) False

We shall prove that for any sequence $\{x_n\}$, if it has a bounded subsequence, then it has a convergent subsequence. Suppose $\{x_n\}$ has a bounded subsequence $\{x_{n_k}\}$. Then $\{x_{n_k}\}$ is also a bounded sequence and so, by the Bolzano-Weierstrass Theorem, contains a convergent subsequence $\{x_{n_{k_t}}\}$. But the sub-subsequence $\{x_{n_{k_t}}\}$ is also a subsequence of the original sequence $\{x_n\}$, it follows that $\{x_n\}$ must contain a convergent subsequence.

(c) False

(See also *Example 3.3*) Consider the series

$$\sum_{n=1}^{\infty} (-1)^{n+1} = 1 - 1 + 1 - 1 + 1 - 1 + \cdots$$

Its sequence of partial sums is given by

$$\{s_k\} = \{1, 0, 1, 0, 1, 0, \dots\}$$

Clearly, this sequence is bounded (as $|s_k| \leq 1$ for any $k \in \mathbb{N}$) and is divergent (as it contains two subsequences converging to two different limits). It follows that the series diverges.

(d) True

For any sequence $\{a_n\}$ with $a_n > 0$ for all $n \in \mathbb{N}$, if $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$ by *Theorem 3.7*. Then we have

$$\lim_{n\to\infty} \frac{1}{a_n} = \infty$$

(see also *Assignment 2, Q.2*). This means

$$\lim_{n\to\infty} \frac{1}{a_n} \neq 0$$

and hence, by *Theorem 3.7* again, we conclude that $\sum_{n=1}^{\infty} \frac{1}{a_n}$ diverges.